

Yichao Cai, PhD Candidate in Computer Science

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RESEARCH PROFILE I study how **training objectives** shape what machine-learning models learn, and when good predictive performance guarantees **recovery of underlying structure**. My research combines mathematical theory with controlled experiments, focusing on causal representation learning, multimodal learning, and foundation models.

RESEARCH INTERESTS Multimodal learning; vision-language models; identifiable representation learning; learning objectives for foundation models.

EDUCATION

PhD Candidate in Computer Science Sep 2023–Present
Adelaide University (Expected completion: June 2027)
Supervisors: Prof. Javen Qinfeng Shi (principal) and Dr Zhen Zhang
Publications: **6 papers** across ICML, NeurIPS, ICLR, and ECCV; **3 as first or co-first author**, including a **NeurIPS Spotlight**.

M.Sc. in Instrument Science and Technology Sep 2016–Jun 2019
Wuhan University of Technology

B.Eng. in Measurement and Control Technology and Instrument Sep 2012–Jun 2016
Wuhan University of Technology

- PUBLICATIONS**
- [1] [The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence.](#)
Cai, Y., Zhang, Z., Liu, Y., and Shi, J. Q.
International Conference on Machine Learning (**ICML**), 2026.
 - [2] [On the Value of Cross-Modal Misalignment in Multimodal Representation Learning.](#)
Cai, Y.^{*}, Liu, Y.^{*}, Gao, E., Jiang, T., Zhang, Z., van den Hengel, A., and Shi, J. Q.
Advances in Neural Information Processing Systems (**NeurIPS**), 2025. **Spotlight**.
^{*}Equal contribution.
 - [3] [CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts.](#)
Cai, Y., Liu, Y., Zhang, Z., and Shi, J. Q.
European Conference on Computer Vision (**ECCV**), 2024.
 - [4] [I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data?](#)
Liu, Y., Gong, D., **Cai, Y.**, Gao, E., Zhang, Z., Huang, B., Gong, M., van den Hengel, A., and Shi, J. Q.
International Conference on Learning Representations (**ICLR**), 2026.
 - [5] [What Makes a Representation Good for Single-Cell Perturbation Prediction?](#)
Jiang, W., Liu, Y., **Cai, Y.**, Gao, E., Dong, J., Abbasnejad, E., Yao, L., and Shi, J. Q.
International Conference on Machine Learning (**ICML**), 2026.
 - [6] [Boundary Embedding Shaping with Adaptive Contrastive Learning for Graph Structural Disentanglement.](#)
Chen, J., Yin, Z., **Cai, Y.**, Liu, Y., Zhang, Z., Gong, D., and Shi, J. Q.
International Conference on Machine Learning (**ICML**), 2026.

PREPRINTS	[1] An Identifiability Theory of Masked Prediction: Mode Blindness and Mask Schedules. Cai, Y. and Shi, J. Q. arXiv:2608.01383, 2026.	
INVITED TALKS	Understanding Multimodal Contrastive Learning through Two Lenses Online invited talk, <i>Vision & Sensing Lab, Technion</i> Online invited talk, <i>EML, Helmholtz Munich</i>	Jul 2026
HONORS & AWARDS	D. R. Stranks Travelling Fellowship NeurIPS Scholar Award University of Adelaide Research Scholarship Outstanding Graduate, Wuhan University of Technology	Jun 2026 Oct 2025 Sep 2023 Jun 2019
STUDENT MENTORING	Master’s Research Project Co-Mentor EML, Helmholtz Munich Co-mentoring a master’s research project on masked prediction and representation learning.	2026–Present
TEACHING EXPERIENCE	<i>Adelaide University</i> (Formerly the University of Adelaide) Teaching Assistant, <i>Programming for Artificial Intelligence</i> Teaching Assistant, <i>Neural Networks and Deep Learning</i> Guest Lecturer & Head Tutor, <i>Statistical Machine Learning</i> Delivered a guest lecture on vision-language modeling and coordinated the tutor team. Teaching Assistant, <i>Using Machine Learning Tools</i> Developed automated scripts for efficient and consistent assignment marking. Teaching Assistant, <i>Concepts in AI and Machine Learning</i>	Semester 2, 2026 Semester 1, 2026 Semester 2, 2025 Trimester 2, 2025 Semester 1, 2025
ACADEMIC SERVICE	Conference reviewer: AAAI 2027; NeurIPS 2026; ICML 2026; ICLR 2026, 2027. Journal reviewer: Transactions on Machine Learning Research (TMLR).	
VISITING RESEARCH	Visiting Student Researcher <i>California PATH, UC Berkeley</i> Conducted research on visual-perception algorithms for autonomous driving, including model development and experimental evaluation.	May 2018–Sep 2018
TECHNICAL EXPERTISE	Methods: identifiability analysis, latent-variable modeling, population-objective analysis, representation geometry, probing, controlled experimentation, and benchmark design. Software: Python, PyTorch, C/C++, Linux, and Git. Languages: Mandarin Chinese (native); English (professional working proficiency).	